

Binary Cross Entropy Loss function:

$$L = - [p \log(a) + (1-p) \log(1-a)]$$

We are required to find $\frac{dL}{da}$

$$\frac{dL}{da} = - \left[\frac{d}{da} p \cdot \log(a) + \left(\frac{d}{da} (1-p) \right) \cdot \log(1-a) \right]$$

$$= - \left[p \cdot \frac{d}{da} (\log(a)) + (1-p) \cdot \frac{d}{da} (\log(1-a)) \right]$$

[Since p and $(1-p)$ are constant]

$$= - \left[p \cdot \frac{1}{a} + (1-p) \cdot \left(\frac{1}{1-a} \right) \cdot (-1) \right]$$

$$= - \frac{p}{a} + \frac{(1-p)}{(1-a)}$$

$$= \frac{-p(1-a) + a(1-p)}{a(1-a)}$$

$$= \frac{-p + ap + a - ap}{a(1-a)}$$

$$\frac{dL}{da} = \frac{a-p}{a(1-a)}$$

hence, $\frac{dL}{da} = \frac{a-p}{a(1-a)}$